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NORMAN D WATKINS, BERNARD M GUNN, AJOY K BAKSI, DEREK YORK and JAMES ADE-HALL

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NORMAN D. WATKINS — *Graduate School of Oceanography, Narragansett Marine Laboratory, University of Rhode Island, Kingston, Rhode Island 02881*

BERNARD M. GUNN — *Department of Geology, University of Montreal, Montreal, Quebec, Canada*
AJOY K. BAKSI — *Department of Geological Sciences, Queens University, Kingston, Ontario, Canada.*
DEREK YORK — *Geophysics Division, Department of Physics, University of Toronto, Toronto 5, Ontario, Canada*

JAMES ADE-HALL — *Department of Geology, Dalhousie University, Halifax, Nova Scotia, Canada*

Paleomagnetism, Geochemistry, and Potassium-Argon Ages of the Rio Grande de Santiago Volcanics, Central Mexico

ABSTRACT

Seventy-two oriented cores of 2.5 cm diameter have been drilled from seven bodies comprising part of a sequence of ignimbrite sheets interdigitated with basalts and basaltic andesites, in the Grand Canyon of the Rio Grande de Santiago of the Mexican Cordilleran Volcanic Province. Field observations and whole-rock analyses reveal two groups: an olivine basalt–high alumina basalt–basaltic andesite series, and a sequence of rhyolitic ignimbrite sheets, with some rhyolitic and obsidian lava flows. Crystal separation, partial melting of oceanic tholeiite, and simple mixing of basalt and oceanic sediments or granodiorite cannot acceptably explain the observed fractionation trends. The favored origin is a partial melting of a second or third stage product, such as a basalt or amphibolite enriched in K, Rb, Th, and Pb, relative to oceanic tholeiite. The observed fractionation trends are parallel to those of the andesite-dacites of the Mexico City region, but the Rio Grande de Santiago rocks have a systematically higher K_2O/SiO_2 ratio, which can be explained by a greater magma generation depth.

The paleomagnetic polarity sequence with decreasing age is R,R,N,R,R,N,N, providing easily measurable markers of high potential stratigraphic value. The mean virtual geomagnetic pole is not significantly different from the present spin axis. The potassium-argon analyses show that the sampled section contains two distinct volcanic episodes, from approximately t = 4.6 to 5.5. m.y. and from approximately t = 8.7 to 9.5 m.y. The polarities recorded by the youngest volcanic cycle are consistent with the predicted polarity time scale of Heirtzler and others (1968), while the older cycle polarities can be explained by minor modification of the predicted polarity scale, to include a short period of reversed polarity between 9.0 and 9.5 m.y., within the normal polarity anomaly 5 (t = 8.79 to 9.94 m.y.) of the time scale.

INTRODUCTION

Within Mexico, volcanic lavas and pyroclastics cover more than one-third of the total land area. The two principal volcanic regions are the Cordilleran Province, and the Rio Lerma Province. The former extends from the United States–Mexico border to south of Guadalajara, and from the Pacific coast to Torreon in the Central Plateau. It is dominated by vast ignimbrite sheets. The Rio Lerma Province extends south and east of Guadalajara, and is sometimes called the Central Mexico Volcanic Belt. It includes the active andesitic strato-volcanoes Volcan Colima and Volcan Popacatepetl, as well as many hundreds of extinct andesitic cones of early Miocene to Recent age. The two provinces intersect in the Guadalajara area with the andesitic stratovolcanoes overlying and probably interdigitated with the older Cordilleran Province ignimbrites. As far as we know, no quantitative analyses of any geological or geophysical parameters of this pivotal area have been published.

In common with other volcanic regions, stratigraphic studies in central Mexico are handicapped by the general lack of distinctive fossiliferous sedimentary horizons. It is the intention of this paper to present the paleomagnetism, geochemistry, and potassium-argon ages of the volcanic sequence exposed in the

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canyon of the Rio Grande de Santiago (Fig. 1), immediately north of Guadalajara. Since there is now no doubt that the geomagnetic polarity has changed many times during the Tertiary epoch, the paleomagnetic polarities which we shall present can be considered as readily obtainable nondiachronous field mapping units, as pioneered by Einarsson (1957) for lava sequences in Iceland. Watkins and others (1966) have summarized methods which can be employed to measure the magnetic polarity of volcanic units in the field. When combined with potassium-argon data, the magnetic polarities can obviously provide a quantitative dual stratigraphic approach which can, in the case of the Guadalajara volcanics, probably be continued a considerable distance to the north. Such data may also contribute to finer definition of the

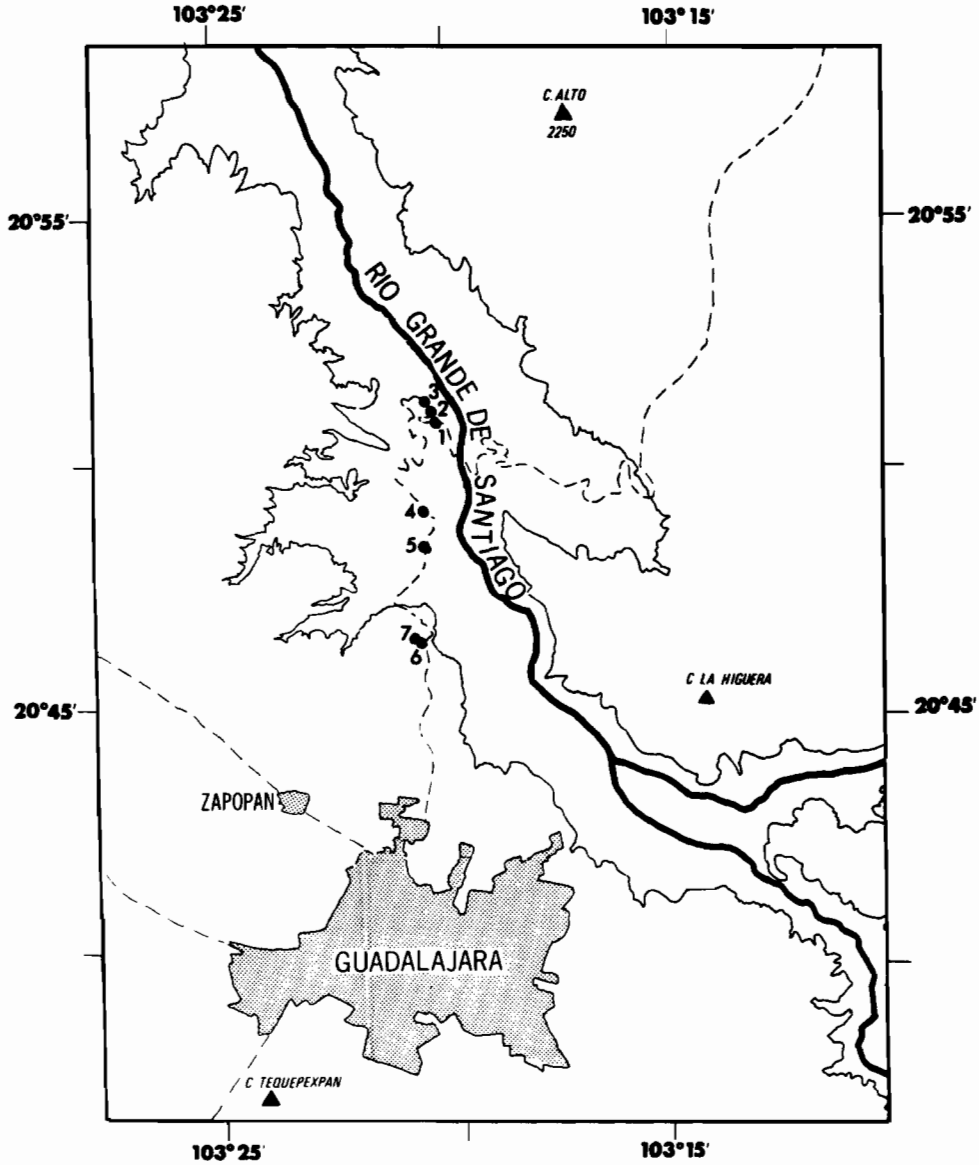


Figure 1. Location of sampled bodies (1 to 7) in the canyon of the Rio Grande de Santiago. Topographic contour is 1500 m above sea level.

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geomagnetic polarity history (Cox, 1969) which is relevant to detailed resolution of the crustal-spreading hypothesis (Heirtzler and others, 1968). Finally, geochemical analyses and petrographic descriptions of the Guadalajara volcanics are relevant to further delineation of stratigraphic units, as well as regional upper mantle processes. For this particular part of Central America, hypotheses involving great crustal movements are inevitable in reconstruction of the North Atlantic, and therefore geochemical data may contribute to placing limits on such models.

GEOLOGY OF THE AREA

The Rio Lerma Volcanic Province of southern Mexico is seismically very active and features numerous faults and grabens which trend east-west. The igneous rocks of the area are mainly andesite-dacite, with clusters of obsidian domes and rare basalt flows. Small cones of the Paracutin type having a composition of basaltic andesite or olivine basalt, occur in rows along east-west fractures between the main andesite cones. Near Mexico City, the principal stratovolcanoes lie along north-south fractures. Ignimbrites are very rare. In contrast, the Cordilleran Province is a region of low but persistent seismicity with very thick, flat-lying ignimbrite sheets and crystal-vitric tuffs or lithic tuffs dominant. In upper sections, the ignimbrites are intercalated by flows of high-alumina basalt, olivine basalt, and basaltic andesite.

The whole province is underlain by an undulating erosion surface cut in Cretaceous carbonates, which near Mazatlan is covered with folded and deformed andesite breccias. These Cretaceous rocks are intruded by diorite, granodiorite, and syenite batholiths.

The Grand Canyon of the Rio Grande de Santiago (Fig. 1) exposes a section typical of the upper part of the Cordilleran Province. As far as we know, the only available published large-scale geological sketch map of any part of the area is that which accompanied the literature for excursion A16 at the 29th International Geological Congress in Mexico City. The recently published 1:2,000,000 geological map of Mexico shows the area around Guadalajara as "Pliocene" in age.

Figure 1 shows the locations of the numbered sample sites on the relatively new road cut in the canyon north of Guadalajara. Figure 2 shows a diagrammatic cross section of the sample area. A thick basal rhyolitic ignimbrite is succeeded by two thin lavas and some basic

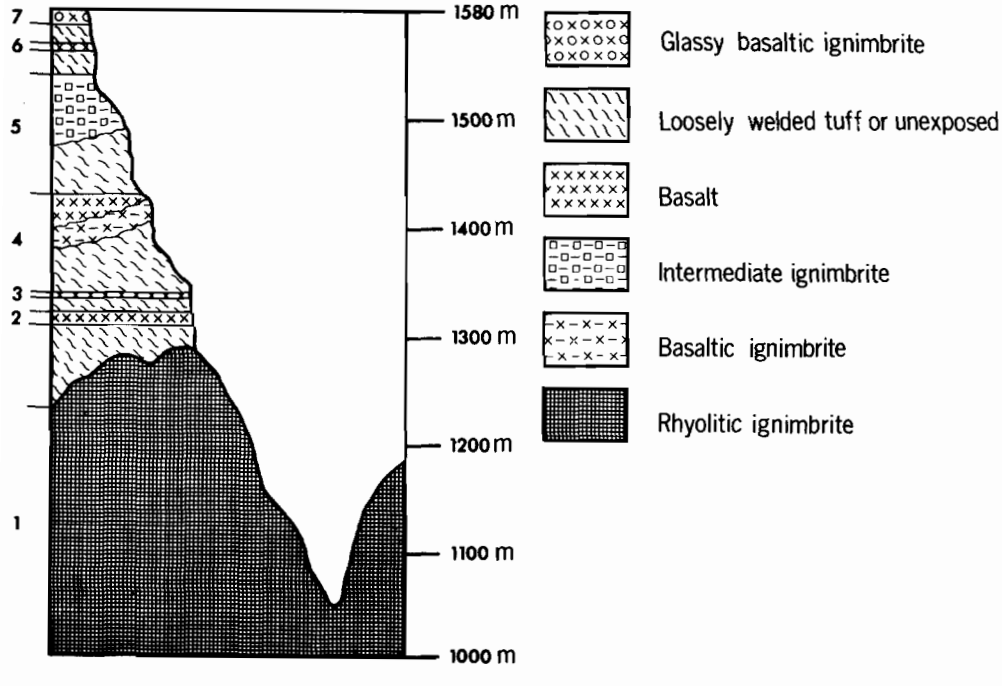


Figure 2. Diagrammatic cross section of southwestern side of the Rio Grande de Santiago, showing location of the seven sampled bodies, and general petrological types. Elevation is in meters above sea level.

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ignimbrites, another lava, and a prominent capping basaltic ignimbrite sheet. Covered intervals exist, implying that additional discrete bodies may be present. Some of the units are undoubtedly very widespread. The apparent lack of lavas of dacite composition in the Santiago Canyon section may be accidental, because the high viscosity of dacite restricts its potential areal extent.

EXPERIMENTAL METHODS

Field

A total of 72 geographically oriented cores of average length 12 cm and 2.5 cm diameter were drilled, using a gasoline-powered drill, from the seven successive bodies (Fig. 2). In addition, larger unoriented samples were taken.

Geochemistry

Thirteen whole-rock analyses of the major oxides and trace elements have been made on drill cores and hand samples, using x-ray fluorescence methods. Powders were fused in a 2 to 1 mix with lithium tetraborate flux, and analyzed, using a Phillips 1220 KV semiautomatic spectrometer. A Tectron atomic absorption spectrometer was used for sodium analyses; phosphorus concentration was measured by colorimetry; and ferrous iron by titration with permanganate. Further details of the analytical methods, precision, and data reduction techniques are presented in Gunn and Watkins (1969, 1970). The analyses were supplemented by conventional transmitted light thin-section microscopy.

Paleomagnetism

Specimens of 2.2 cm length were cut from each core for paleomagnetic analysis. The remanent magnetic direction and intensity was measured using a 5 c/s spinner magnetometer (Foster, 1966) before and after demagnetization in alternating magnetic fields of 100, 200, and 300 oersteds (McElhinny, 1966) except in unit 6, where alternating magnetic fields of up to 1200 oersteds were applied. Conventional statistical analyses (Fisher, 1953) were made on all data, and the mean direction of remanent magnetism computed, using the least dispersion method (Irving and others, 1961) on all demagnetized material. The low-field susceptibility was measured, using a 300 c/s bridge (Collinson and others, 1963).

Polished thin sections were made from one core in each body for opaque mineralogical analysis, using a Reichert Zetopan-pol reflection microscope, and oil immersion. The magnetite oxidation number (M) and magnetite granulation number (G) were measured in each case. These parameters are defined by Ade-Hall and others (1968). M is calculated following identification of the high temperature oxidation index of the titanomagnetites present on a I to VI scale (corresponding to homogenous titanomagnetite to hematite-pseudobrookite intergrowths, with intermediate levels dependent on the texture and stages present) and application of the equation:

$$M = \sum_{n=1}^{6} (n \times \text{percentage of total titanomagnetite area covered by index } n).$$

G describes the relative amount of a granular alteration production present in the titanomagnetites. The product, first described by Ade-Hall and others (1968), is a form of ferrirutile. G is defined by $G = (P/20 + 1)$ where P is the mean area percentage of granulation for 50 titanomagnetite grains per specimen. The mean area per titanomagnetite grain is estimated to ± 10 percent. The definition of G is such that it varies from granulation absent (1.0) to granulation throughout (6.0) for direct comparison with M.

The Curie point of one specimen from each body was measured using a modified Chevallier balance. The specimen is heated in air sufficiently fast to inhibit the effect of chemical changes during the heating cycle.

Potassium-Argon Dating

Four of the flows sampled were ignimbrites (1, 4, 5, and 7). These rocks are not normally dated by the whole-rock technique. Evernden and Curtis (1965) recommended K-Ar dating of mineral separates from such rock types. Limited whole-rock dating of these materials has nevertheless been attempted, because of the stratigraphic control available. The other units are basalts and have also been dated by the whole-rock technique.

Samples were prepared for potassium and argon analysis following Baksi and others (1967, p. 6302, method two). The potassium analyses were carried out using lithium internal standard and a sodium buffer with an I.L. 143 flame photometer. Most of the rocks were quite homogeneous in potassium content and the average of the concentrations for the two ends of

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a core was used. Argon analyses were made in at least duplicate for each flow using an MS 10 mass spectrometer.

RESULTS

Petrography

Thin sections from the seven successive units (Fig. 2) are described:

Unit 1 (Core 6). This rhyolitic ignimbrite sheet exhibits classical welded flattened tri-cuspate glass shards enclosing 5 mm fragments of crystalline rhyolite. No ferromagnesian minerals are seen, but fine opaque minerals are present. A few brown hornblende grains may be seen in thin section.

Units 2 (Core 7) and 3 (Core 5). These are olivine basalts. They contain 5 to 12 percent of rounded to subhedral iddingsitized or serpentinized olivines which are often grouped together. Plagioclase laths range commonly from 0.2 mm to 2 mm, and rarely as much as 4 mm, and feature seriate texture. Clinopyroxene is purple in color and forms subophitic intergrowths with the finer plagioclase. Titanomagnetic and apatite rods, and interstitial green serpentine are common.

Unit 4 (Cores 3 and 10). This is a fine-grained basaltic to ignimbritic body. Fluxion-oriented 0.2 mm-long plagioclase makes up 60 percent of the rock. Subparallel leucocratic patches emphasize the banded character. Patches of chlorite in the groundmass may be altered glass. Some amygdales are of radially arranged calcite.

Unit 5 (Core 4). In outcrop, this andesite ignimbrite grades laterally into what appears to be a fine-grained holocrystalline andesite. The unit is highly oxidized. In thin section, hematized pyroxenes, hematized titanomagnetites, and iddingsitized olivines are common. The groundmass is dominantly 0.1 to 0.2 mm plagioclase, and granular pyroxene. Glass is rare, but lenticular when observed.

Unit 6 (Cores 7 and 8). This is a holocrystalline basalt. The groundmass is clinopyroxene, orthopyroxene, and plagioclase. Porphyritic plagioclases (up to 10 mm long), and irregular clinopyroxene and orthopyroxenes are characteristic.

Unit 7 (Core 6). This is an unusual rock, identified by us as a glassy basaltic ignimbrite in outcrop. In thin sections, it consists of 1 to 3 mm broken plagioclase, orthopyroxene, amphibole, augite, biotite, and iddingsitized olivine, together with fragments of rhyolite, andesite, basalt, and obsidian in an opaque glass matrix.

Geochemistry

The results of the 13 chemical analyses are presented in Figure 3 as functions of the differential index of Thornton and Tuttle (1960), which is the sum of the normative albite + nepheline + leucite + quartz. Means are included in Table 1.

Paleomagnetism

Table 2A contains the paleomagnetic results. Simple normal or reverse polarity bodies are defined. Computed final mean directions are given on Figure 4a and the corresponding virtual geomagnetic pole positions are plotted in Figure 4b. The mean direction of remanent magnetism for the entire survey is given in Table 2B. The opaque mineralogical and Curie point data are given in Table 3.

Potassium-Argon Results

The results are given in Table 4.

DISCUSSION

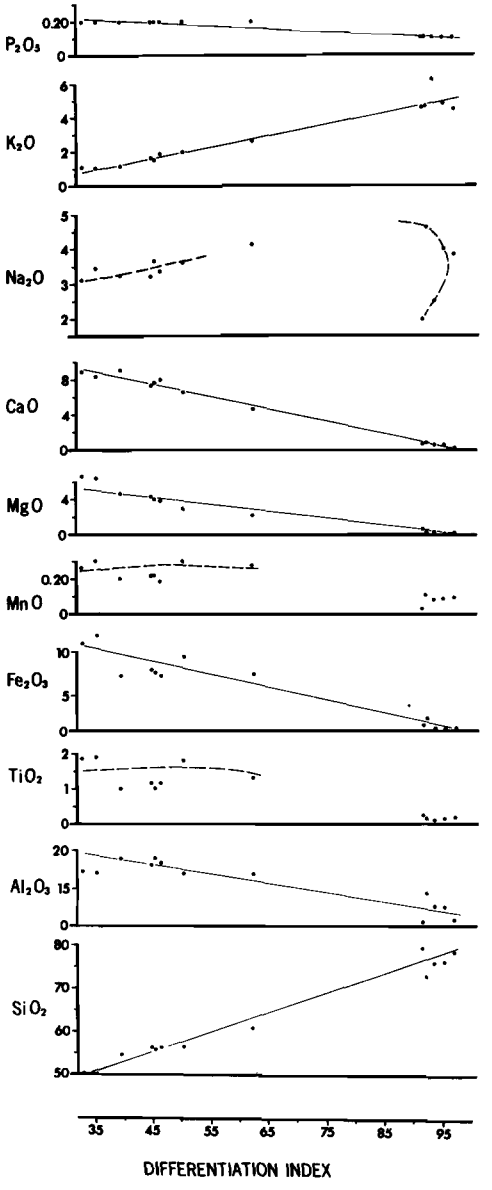
Geochemistry

The Santiago Canyon volcanics are chemically bimodal. The two series are an olivine basalt to high alumina basalt to basaltic andesite, and a rhyolitic ignimbrite to rhyolitic obsidian. From the variations diagrams (Fig. 3), we see that although a gap is present, the basaltic rocks are distributed quite broadly along trends directed toward the rhyolite-ignimbrites. Rocks of andesite-dacite composition, which make up the bulk of the lavas in the Rio Lerma Province, fall into this compositional gap (Gunn and Mooser, 1971).

The projection of the basaltic andesites toward the ignimbrites can be interpreted to be a result of partial contamination of basalt. Taylor (1969) has shown that the composition of andesite is incompatible with an origin involving mixing of basalt with oceanic clays or granodiorite. If both rhyolitic and basaltic magma are being produced simultaneously in the same region, some hybridization will almost certainly occur. The characteristics of one such

[1]The practice of applying a prefix, other than rhyolitic, to the term ignimbrite to describe a rock unit has frequently been used elsewhere. For example, Walker (1962) describes basaltic ignimbrites of eastern Iceland, and Schmincke (1969) describes the trachyphonolitic ignimbrites of Gran Canaria.

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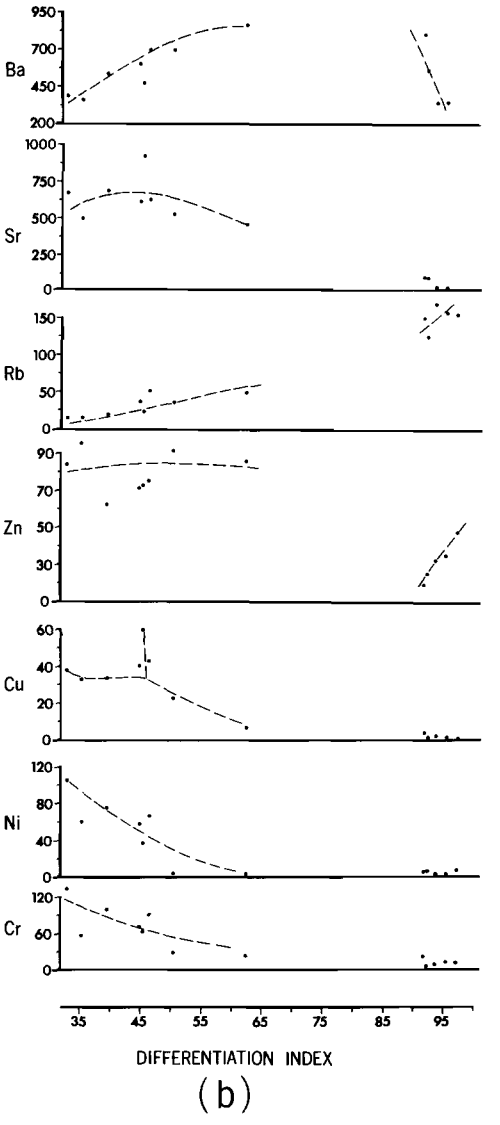


Figure 3. a. Concentrations of major oxides (in weight percent); and b. Trace elements (in ppm) as a function of the Thornton-Tuttle (1960) differentiation index (normative Ab + Ne + Le + Qz) for the 13 analyzed specimens. Trend lines are added to emphasize observed trends, but are not intended to represent proven real trends in all cases.

hybrid in Iceland has been investigated by Gunn and Watkins (1969). The variation diagrams for Ba and Na for the Santiago Canyon rocks show however, that contamination would have to occur between basalt and a rock more basic than any of the rhyolites present in the area. The only material of suitable composition (especially in Na and Ba) known to us is a granodiorite under lying the volcanics in the Mazatlan area. For all the elements analyzed (B.M. Gunn, unpub. data) a few percent of this granodiorite mixed with the basalt could produce the variations observed in the Santiago Canyon. The contamination would, however, have to be a total and not a partial melt of granodiorite.

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Andesitic magmas can also be generated by the partial melting of quartz eclogite. The eclogite may be formed, for example, by oceanic crust (consisting mainly of oceanic tholeiite) underthrusting a continental craton (Green and Ringwood, 1968). The first fractions of such a partial melt would be the most silicic and only the residual fractions would have the composition of the Santiago Canyon basalts. The sampled basalts, however, are unlikely to be residual, as all contain at least 1 percent K_2O. Regional tectonic analyses will need to incorporate these limitations.

Crystal separation may cause fractionation trends. The normative basalt tetrahedron for the Santiago Canyon Volcanics (Fig. 5) shows

TABLE 1. GEOCHEMICAL ANALYSES

A. Major oxides (in weight percent)

Oxide	Average for basalt to andesite	Average rhyolite
SiO_2	55.00	76.50
Al_2O_3	17.46	12.58
TiO_2	1.40	0.16
Σ Fe	8.75	1.57
MnO	0.121	0.039
MgO	4.37	0.231
CaO	7.59	0.538
Na_2O	3.48	3.401
K_2O	1.62	4.92
P_2O_5	0.2	0.10

B. Trace elements (in ppm)

Element	Average for basalt to andesite	Average rhyolite
Cr	70.5	11.00
Ni	52.2	6.00
Cu	34.7	2.30
Zn	79.8	31.70
Rb	29.3	150.00
Sr	627.0	58.00
Ba	579.0	455.00

C. Selected element ratios

Ratio	Value for basalts and andesites	Correlation coefficient	Value for rhyolites	Correlation coefficient
K / Rb	460.0	0.89	272	0.61
K / Ba	23.	0.96	90	0.25
Ca / Sr	118.	0.46	66	0.68
Mg / Cr	374.0	0.70	127.0	0.74
Mg / Ni	505.0	0.84	232.0	0.11
Cr / Ni	1.4	0.96	1.8	0.07
Fe / TiO_2	6.2	0.92	9.8	0.09
Fe / MnO	71.0	0.72	40.0	0.23

TABLE 2. PALEOMAGNETIC DATA

A. Each Unit

No.	N	D	I	Ko	K100	K200	K300	Kmax	J × 10^{-5}	σJ × 10^{-6}	φ'	θ'
7	10	356.3	+09.2	24.2	77.4	152.2	169.7	203.8	108.0	233.2	89.7	+73.5
6	6	330.3	+46.9	1.4	1.3	3.9	3.0	69.2	35.0	134.0	187.7	+62.0
5	8	186.1	-27.1	71.7	344.4	4.5	431.3	561.9	41.0	271.3	213.5	-81.3
4	17	178.7	-34.7	145.5	14.4	101.7	78.2	122.3	76.0	523.0	318.9	-86.4
3	8	358.6	+45.0	5.9	15.8	7.6	27.1	33.5	11.0	88.8	244.1	+84.1
2	8	165.1	-21.0	9.3	38.6	94.7	23.9	284.5	15.0	82.7	314.2	-72.6
1	12	144.8	-58.1	81.6	20.2	154.4	47.3	167.5	5.0	9.9	25.2	-54.9

B. All Units Combined

B	N	D	I	K	α95	dp	dm	φ'	θ'
7	69	349.8	+34.8	16.9	12.9	8.5	14.8	158.8	+80.3

In Part A: No. = number of body (Fig. 2); N = total number of cores; D and I = declination and inclination of remanent magnetism, following demagnetization and minimum dispersion computation (Irving and others, 1961); D is in degrees of north, I is with respect to horizontal (+ is below and - is above horizontal); K = N - 1/N-R where R = total resultant vector applying unit vector per core suffix = size of rock alternating magnetic field applied (100 to 300 oersteds) or max (minimum dispersion computation which includes demagnetization up to 1200 oes in unit 6 (*see* Fig. 6); J = intensity of magnetization in emugm^{-1}; σJ = standard deviation of J; φ' and θ' = longitude and latitude of virtual geomagnetic pole; φ' is degrees east, θ' is degrees north (+) or south (-).

In Part B: as in Part A, except B = total number of bodies, K = B-1/B-R, where R = unit vector per body; α95 = semi-vertical angle of 95 percent confidence cone; dp and dm = angular length of semi-axes of 95 percent confidence oval, along and normal to great circle between sample site and VGP.

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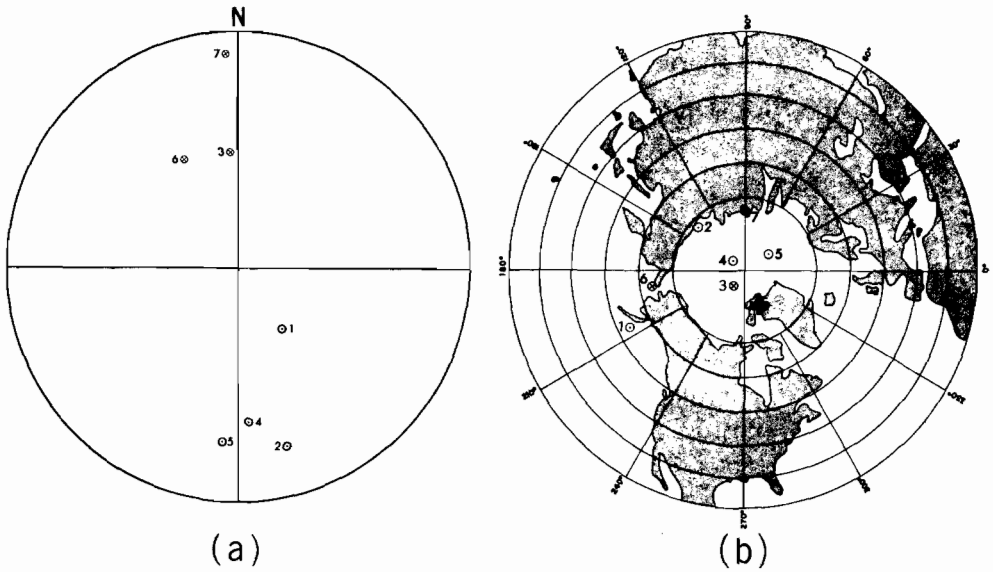


Figure 4. a. Mean directions of remanent magnetism in each unit. ⊗ = lower hemisphere. ⊙ = upper hemisphere. b. Corresponding positions of virtual geomagnetic poles for each unit. N = present geomagnetic pole position. ⊙ = reversed polarity. ⊗ = normal polarity. Reversed polarity data are transposed to northern hemisphere.

TABLE 3. RESULTS OF DETAILED MEASUREMENTS ON SELECTED SPECIMENS

Specimen number	M	G	Tc
7-7	1.02	1.04	510
6-7	1.16	1.00	280,480
5-8	5.15	1.00	400,540
4-2	2.28	1.92	300,540
3-3	2.56	1.00	360,540
2-1	1.19	1.03	320,530
1-1	6.00	1.00	too weak to measure

Specimen number: First digit is body number (Fig. 2), second digit is core number; M = titanomagnetite oxidation number (*see* text); G = titanomagnetite granulation number (*see* text); Tc = Curie point(s) in degrees centigrade.

no trend either toward the olivine apex or away from it, meaning that fractionation due to olivine addition or removal can only be present to a very minor degree. For a basalt liquid to fractionate along the invariant line shown, roughly equal amounts of plagioclase-clinopyroxene and olivine would first have to be removed until the Ab-Di-Hy plane was reached, at which time hypersthene would take the place of olivine. Such regular removal from a melt of three crystal phases of markedly different densities is considered impossible. There is (as far as we know), in any case, a complete absence of any cumulative rocks associated with concentration of any one phase.

We conclude that these samples are probably

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TABLE 4. POTASSIUM-ARGON ANALYSES

Sample No.	K content (%)	Atmos. Ar 40 (%)	Ar 40* × 10^7 (ccNTP/gm.)	Age (m.y.)	Mean age and estim. stnd. dev. (m.y.)	Polarity
7 - 1	2.83	64.0	5.42	4.78	4.71 ± 0.05	N
		67.4	5.11	4.52		
7 - 4	2.87	69.3	5.48	4.78		
		69.0	5.44	4.74		
6 - 2	1.50	82.2	2.79	4.65	4.65 ± 0.10	N
		80.9	2.78	4.64		
5 - 1	0.95	79.6	2.05	5.40	5.37 ± 0.07	R
		80.6	2.01	5.28		
5 - 7	0.92	77.7	2.00	5.43		
		78.3	1.96	5.33		
5 - 2	3.76	40.0	8.03	5.34		
		10.2	8.22	5.46		
4 - 5	1.28	88.3	4.48	8.72	8.75 ± 0.20	R
		86.9	4.50	8.77		
3 - 1	0.76	56.4	2.72	8.91	8.96 ± 0.10	N
3 - 2	0.67	47.2	2.43	9.02		
		47.2	2.43	9.02		
3 - 3	0.68	58.4	2.42	8.90		
2 - 3	0.79	61.3	2.86	9.08	9.27 ± 0.12	R
		74.4	2.95	9.36		
2 - 6	0.71	89.3	2.77	9.29		
		93.2	2.78	9.34		
1 - 1	3.90	63.6	13.9	8.89	8.88 ± 0.12	R
		60.9	14.0	8.98		
1 - 5	4.11	58.5	14.3	8.67		
		50.9	14.8	9.00		

Ar 40* = radiogenic Ar 40.
Constants used: $\lambda\beta = 4.72 \times 10^{-10}$ yr^{-1}; $\lambda e = 5.84 \times 10^{-11}$ yr^{-1}.
K40 = 0.0119% atomic of total K.
Sample number system: first number = body (Figs. 1 and 2); second number = core number. Magnetic polarity (Table 2) added for comparison; N = normal polarity; R = reversed polarity.

partial melting products of a second- or third-stage zone refining product, possibly a basalt or amphibolite already enriched in K, Rb, Ba (and by analogy with the andesites near Mexico City, Th and Pb) relative to oceanic tholeiite. The possibility that they may be a series of partial melts derived from the upper mantle, unmodified significantly by either fractionation or granodiorite contamination, is very low.

Comparison with fractionation trends defined by 64 analyses of the andesites-dacites from the Mexico City region (Gunn and Mooser, 1971) shows that many trends are parallel, but not identical. For example, as Figure 6 shows, the Guadalajara volcanics are systematically lower in MgO, relative to silica. They are also richer in K_2O and Na_2O, and have lower Rb (and hence a higher K/Rb ratio), but have very similar $\Sigma Fe/SiO_2$ and $\Sigma Fe/TiO_2$ ratios. The distinctly higher K_2O/SiO_2 ratio in the Santiago Volcanics could be due to a greater depth of magma generation

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(Dickinson and Hatherton, 1967).

Paleomagnetism

The magnetic stability is high in the three ignimbrite units (nos. 1, 4, and 5). These all have high magnetite oxidation numbers (Table 3). In units 2, 3, and 7, small- to moderate-sized unstable components are evident, but these are readily removed by the demagnetizing treatments. The unstable components in 2 and 7 are associated with low magnetite oxidation numbers. The correlation between stability and titanomagnetite oxidation is well known (Watkins and Haggerty, 1967).

The stability is in all cases sufficiently high that polarities measured in the field would reflect the original ambient geomagnetic polarities: an original reversed polarity would not be masked by unstable components.

The final results (Fig. 4) show virtual geomagnetic poles at latitudes which are sufficiently high for no ambiguities to exist in identifying reversed or normal polarity bodies. A basal reversed polarity rhyolitic ignimbrite is succeeded by reversed and then a normally magnetized basalt, two reversely magnetized ignimbrites, a normal polarity basalt, and a normally magnetized capping basaltic ignimbrite. These polarity differences are clearly very convenient stratigraphic markers.

The mean virtual geomagnetic pole position is at 158.8° E., 80.3° N. At the 95 percent confidence level, this is not significantly different from an axial dipole.

Opaque mineralogical observations (Table 3) show that the titanomagnetites in the ignimbrites (with the exception of the obviously quenched capping glassy basaltic ignimbrite) are all higher oxidation types. The oldest basalt (Fig. 2, unit 2) and the two basaltic ignimbrites (units 4 and 7) display measurable "granulation," which has been known to be associated with regional hydrothermal alteration in zeolite facies conditions resulting from burial by young lava flows (Ade-Hall and others, 1968). One of the present authors (Ade-Hall) believes that this may indicate the loss of a considerable cover by erosion, the actual thickness lost depending on the regional temperature gradient at the time of formation. This author also suggests that the Curie point data (Table 3) are consistent with considerable loss of younger material; the low Tc values for the specimens from units 2, 3, 4, and 6, ranging from 280°C to 360°C, can be interpreted to be consistent with the removal of one kilometer or more of younger material (Ade-Hall and others, 1970). This interpretation is not advocated at this time, however, by the other authors, who believe that there is no evidence to show that the glassy capping basaltic ignimbrite was buried in the

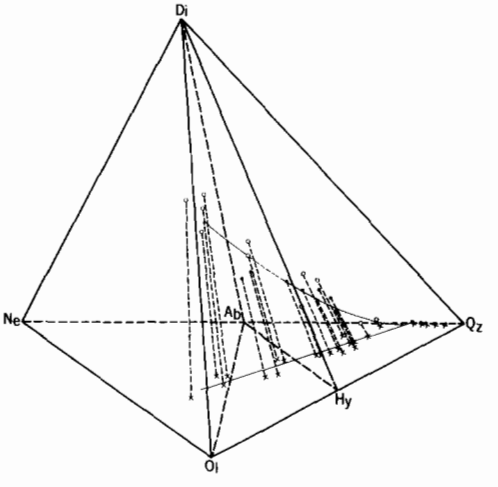


Figure 5. Three-dimensional basalt tetrahedron for the Rio Grande de Santiago volcanics: solid circles are the normative compositions, in terms of diopside-nepheline-olivine-albite-hypersthene-quartz content. Open circles are the data from other analyses in the Guadalajara vicinity (B.M. Gunn, unpub. data). Crosses are the projection of analyses from the diopside apex to the basal plane.

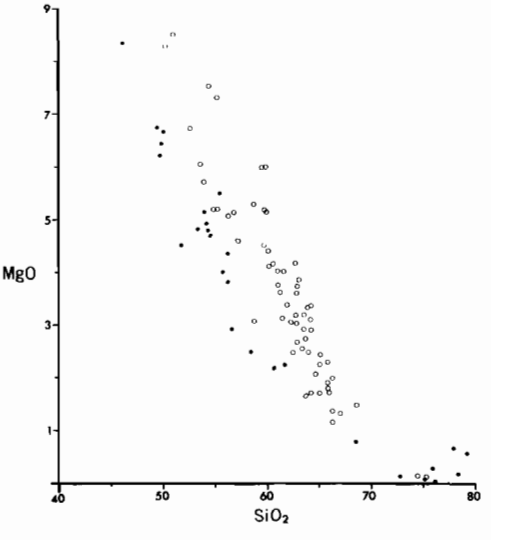


Figure 6. MgO versus SiO_2 plot for the Rio Grande de Santiago volcanics (solid circles) and the Mexico City region (open circles), showing parallel trends but regional differences (Mexico City data *after* Gunn and Mooser, 1971). The Rio Grande de Santiago data include results from materials collected in the Guadalajara area in addition to those sites shown in Figure 1 (B.M. Gunn, unpub. data). Concentrations in weight percent.

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past to any significant depth; it is believed that the occurrence of "granulation" is unlikely to be simply a function of burial depths in ignimbrites.

Potassium-Argon Ages

The ages shown in Table 4 agree well with the stratigraphic sequence with the exception of unit 1 which should be slightly older than its measured K-Ar age. This is almost certainly due to incomplete release of radiogenic argon during fusion of our sample from unit 1, which proved to be unusually difficult to melt completely. The fused samples always contained bubbles and since their upper surface was never horizontal, they would appear to have been highly viscous in the molten state. Webb and McDougall (1967) have reported that sanidines have to be heated to about 1,650°C in order to obtain total release of radiogenic argon and this may be a contributory factor with the result obtained for unit 1.

Although unit 6 seems to be slightly younger than unit 7, the difference in age is not significant within experimental error. Unit 6 was somewhat inhomogeneous in potassium content and the value of this quantity used in the age calculation may therefore be anomalously high.

The age of unit 4 is consistent with its stratigraphic position.

The results on flow five are very encouraging. A fresh sample (Table 4, sample 1), an altered sample (sample 5-7) and a sample with very pronounced lamellar structure (sample 5-2) all yield the same age. This age would seem to be geologically meaningful, as it is difficult to envisage any process by which rocks so different in grain size and K content (Table 4) could all yield the same incorrect age.

The attempt at whole-rock dating of ignimbrites was not totally successful. The age of flow one is almost certainly incorrect. The ages of the other two ignimbrites (units 5 and 7), however, appear to be reliable.

The results show that two periods of igneous activity occurred in the region: between approximately t = 4.6 to 5.5. m.y., and between approximately t = 8.7 to 9.5 m.y. The 3-m.y. break in activity may conceivably be only apparent, since a substantial covered interval coincides with the time break. There are, however, clear petrological and geochemical indications of a time break in the section. In both groups, there is a clear tendency for a decrease in silica content with decreasing age. In the older

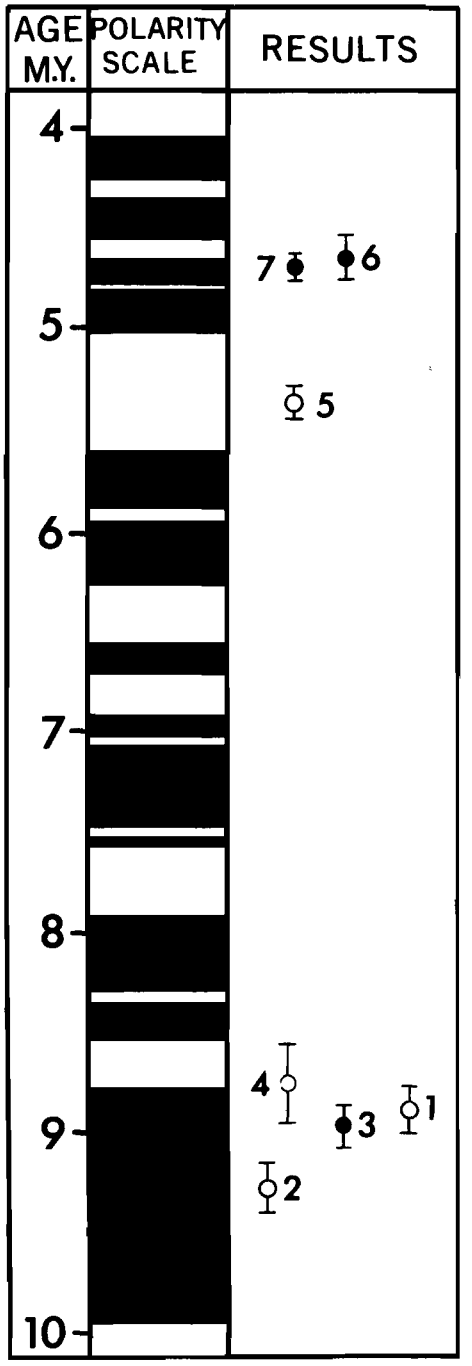


Figure 7. Potassium-argon ages and magnetic polarities of the Rio Grande de Santiago volcanics. Polarity scale is the predicted scale of Heirtzler and others, 1968: black = normal polarity; clear = reversed polarity. Error bars are standard deviations (Table 4). Numbers 1 to 7 correspond to bodies in Figure 2.

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group, rhyolite passes to basalt to basaltic ignimbrite; in the younger group, andesitic ignimbrite passes upward into basalt and basaltic ignimbrite. These may represent corresponding long term magmatic cycles.

Polarity Time Scale

In Figure 7, the polarities and ages are plotted against the predicted polarities of Heirtzler and others (1968). Because of the estimated uncertainties in the potassium-argon ages, the comparison may be of low significance for the older part of the section (Cox and Dalrymple, 1967). When combined with similar data from other parts of the world, however, the results from unit 2 may conceivably be valuable. Such data already exists: Dalrymple and others (1967) report five separate reversely magnetized igneous bodies from the western United States with ages ranging from 8.87 $\pm$ 0.19 to 9.6 $\pm$ 0.2 m.y. Such results are clearly relevant to the reality of the suggested normal polarity anomaly five of the polarity scale of Heirtzler and others (1968), which is suggested to extend from 9.94 to 8.79 m.y. Most recently, Evans (1971) reported two reversed polarity lava flows of 9.1-m.y. age; he suggests the existence of a short reversed polarity event, which would not necessarily be detectable (because of its short duration) in the mid-oceanic magnetic anomaly pattern as used by Heirtzler and others (1968) to construct their predicted polarity history. It will be necessary to devise a statistical method for evaluation of potassium-argon data on a between-laboratory and between-site basis, before the true significance of these data supporting the existence of the possible reversed polarity event at 9.0 to 9.5 m.y. can be evaluated. Such an event is clearly quite conceivable, and now, even probable.

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